5TH INTERNATIONAL CONFERENCE

Smart Grid Synchronized Measurements & *Analytics*

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Benchmarking Dynamic Frequency Estimators for Low-Inertia IBR Grids

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Main claim

- Frequency estimation under grid disturbances is a **multi-objective measurement problem**.
- Rankings change with the objective: frequency RMSE, RoCoF error, latency, CPU cost, and disturbance sensitivity produce different orders.
- IBR-era disturbances make this operational: distorted waveforms and fast controls can turn a measurement artifact into a ride-through or forensic diagnosis problem.
- The result is a reproducible selection protocol: choose by event family, metric, and compute budget.

Core result

A frequency estimator must be evaluated as a deployable measurement system with error, latency, stability, and cost reported separately.

Benchmark at a glance

9+2
estimators

4
families

5
scenarios

grid
search

Main result

RA-EKF improves dynamic tracking and false-trip exposure under IBR composite stress; protection-class frequency measurement needs tests beyond compliance-style cases.

Experimental scope: 9 principal estimators, 2 legacy baselines, 5 stress scenarios, standard and IBR-specific metrics, and a separate expanded sweep for sensitivity analysis.

Evidence map

Stage	Technical content	Evidence shown
Problem	Low-inertia IBR grids compress disturbance timescales and combine stress modes.	Field motivation, research problem, and measurement gap.
Prior work	Standards, PMU measurement, spectral, PLL/SOGI, Kalman, reduced-order, and data-driven estimators.	Benchmark gap and estimator-family taxonomy.
Method	1 MHz waveform generation, 10 kHz estimator stream, 5 scenarios, grid-search tuning, metrics, and CPU timing.	Scenario table, metric equations, RA-EKF mechanism, and traceability.
Results	Standard tests, composite islanding, multi-event trip-risk, CPU, and response dashboards.	Quantitative tables, event plots, Pareto view, and compliance heatmap.
Interpretation	Latency and disturbance sensitivity, state-space advantage, and insufficiency of standard tests.	Findings, recommendation map, and qualification workflow.

Claim chain

Problem: fast IBR stress → Benchmark: 9+2 estimators → Scope: 5 scenarios → RA-EKF: state + gate → Finding: latency + stress

Stress model

- IBR events can combine ramps, phase jumps, harmonics, and ringdown inside one local relay record.
- Scenarios A–C follow IEC/IEEE test philosophy; Scenarios D–E add IBR composite stress.
- Estimators are tuned and evaluated using RMSE, peak error, trip-risk duration, settling, and CPU cost.

Estimator behavior

- RA-EKF reduces false-trip exposure under composite islanding while remaining CPU-feasible.
- Windowed methods are accurate in steady state but carry structural observation delay.
- Compliance tests alone miss phase-discontinuity and multi-event IBR failure modes.

Technical contributions

Scope

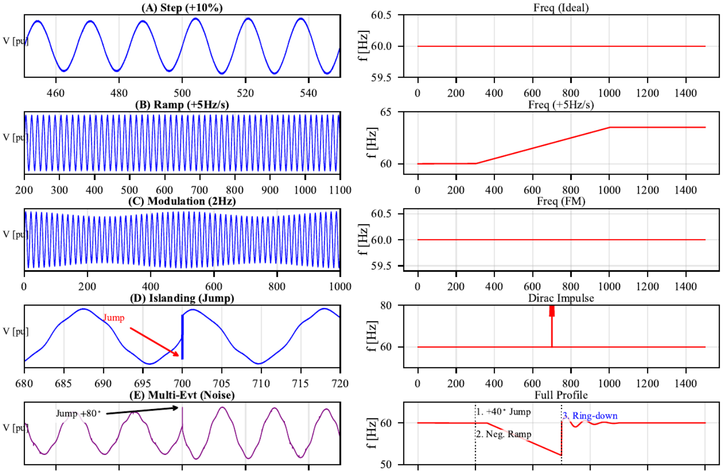
- PMU time-stamp certification;
- a single universal leaderboard;
- hardware latency measurements.

Contributions

- a stress-test benchmark for relay-class frequency estimators;
- an adapted RA-EKF with explicit $\dot{\omega}$ state, covariance scaling, and event gating;
- a metric-complete comparison: RMSE, peak error, T_{trip}, settling, and CPU;
- evidence that standard compliance tests miss IBR failure modes.

The stress-test protocol links RA-EKF design choices to phase-jump false-trip exposure.

Scenario waveforms

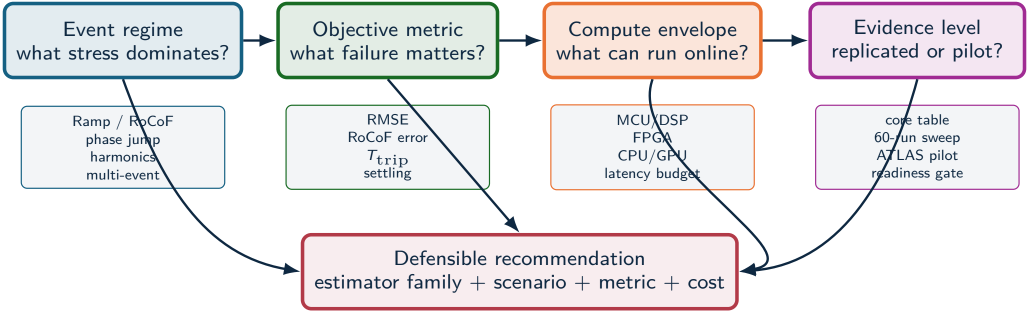


Scenario signatures

Scenario	Signal	Measurement question
Scenario A	10% voltage magnitude step.	Does amplitude change contaminate frequency tracking?
Scenario B	Sustained $+5$ Hz/s frequency ramp.	Can the estimator track RoCoF without excessive lag?
Scenario C	10% AM modulation at 2 Hz.	Does the estimator preserve steady-state accuracy under modulation?
Scenario D	Composite islanding with $+60^\circ$ phase jump, harmonics, interharmonic, and elevated noise.	Does the method recover phase lock without false-trip exposure?
Scenario E	Five-second IBR multi-event sequence with harmonics, jumps, negative ramp, ringdown, and impulsive noise.	Response under stacked stress inside one decision window.

The scenario suite pairs three standard-style references with two IBR-specific composite stress tests.

Estimator selection logic



Example: RA-EKF is recommended for ramp/phase-jump protection metrics under DSP/FPGA-class cost; global RMSE leadership remains scenario-dependent.

Evidence layers: benchmark + ATLAS

Core experiment: 9 principal estimators + 2 legacy baselines, 5 scenarios, standard and IBR-specific metrics

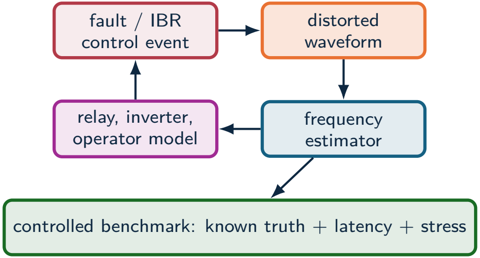
Expanded stress analysis: 16 estimators, 34 scenarios, severity curves, harmonic sweeps, and CPU–accuracy maps

Validation layer: replication depth, fixed policies, artifact manifests, and metric consistency checks

Engineering output: select estimator families by event regime, metric target, and compute envelope

Controlled cases support the main claims; wider sweeps expose severity trends and replication gaps.

From field forensics to estimator tests



$$\widehat{f}_i[k] = \mathcal{E}_i(v[k - L_i : k]; \theta_i)$$

$$\mathbf{J}_i = [\mathrm{RMSE}_f,\ \mathrm{RMSE}_{\dot{f}},\ t_s,\ \tau,\ C_{\mathrm{CPU}},\ p_{\mathrm{invalid}}]$$

- A deployable estimator must stay stable during the event, recover quickly, and expose invalid outputs.
- It must remain useful when the waveform violates its assumptions.

IBR disturbances expose the gap

Event	Observed behavior	Monitoring limitation	Benchmark implication
Blue Cut Fire, CA 2016	Transmission faults caused nearly 1,200 MW of solar PV output loss; NERC linked part of the behavior to near-instantaneous frequency or voltage response.	The critical interval was a distorted, fault-clearing transient on a subsecond timescale.	Test whether estimators confuse waveform distortion with real frequency collapse.
Canyon 2 Fire, CA 2017	Two cleared faults produced 682 MW and 937 MW solar PV reductions.	NERC noted 4 s SCADA could not distinguish momentary cessation from tripping in some cases.	Report latency, settling, invalid samples, event-window behavior, and aggregate error.
Odessa events, TX 2021–2022	More than 1 GW-scale solar reductions followed normally cleared faults far from affected plants.	NERC found controls, protection, and model-fidelity gaps, including effects not captured well by positive-sequence models.	Stress tests must include control-like discontinuities, phase jumps, and model-mismatch regimes.
California BESS, 2022	Normally cleared faults caused abnormal BESS active-power reductions and inverter trips.	Causes included ac overcurrent/overvoltage, dc bus issues, unbalanced currents, and data-quality limits.	Frequency estimators should be evaluated under unbalance and non-sinusoidal contamination.

Source: NERC disturbance reports; NERC IBR disturbance monitoring white paper

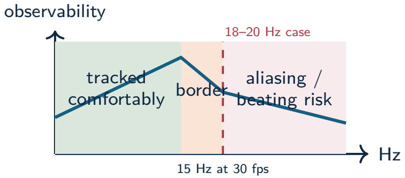
Standard PMU stream limits

Fast fault and ride-through

IBR controls can react within cycles. A low-rate or heavily filtered frequency estimate may miss the subcycle cause while still showing the delayed consequence.

Oscillation near reporting-rate limits

The Kaua'i 2021 event exhibited 18–20 Hz oscillations after a generator trip. A 30 frame/s phasor stream has a 15 Hz Nyquist limit for phasor-time variations, so oscillations near this range can alias or appear as beating.



Harmonics and interharmonics

Standard synchrophasor outputs summarize the fundamental component; harmonic and interharmonic contamination can still dominate estimator failure modes.

Source: Kaua'i 18-20 Hz IBR oscillation analysis; NERC IBR monitoring white paper; IEC/IEEE 60255-118-1

Benchmark requirements from field cases

Field-forensic need

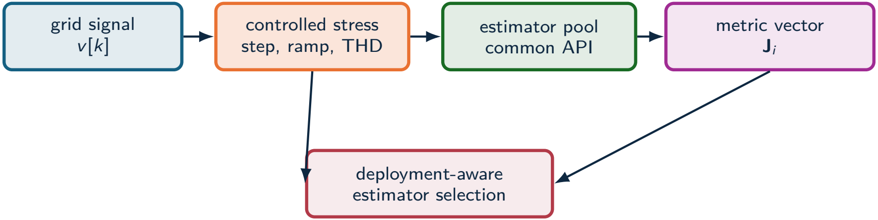
- point-on-wave records above 960 samples/s for triggered fault captures;
- continuous phasor/dynamic disturbance records at ≥ 30 samples/s;
- inverter fault codes, mode changes, and control states;
- traceable event replay for model validation.

Benchmark analogue

- known-truth synthetic scenarios with steps, ramps, phase jumps, THD, and interharmonics;
- estimator outputs logged as time series and scalar summaries;
- latency, CPU, invalid outputs, and settling treated as first-class metrics;
- CSV/JSON artifacts regenerated into figures and tables.

Field oscillography gives realism; controlled synthetic truth isolates which estimator family fails first.

Graphical abstract



Accuracy
Frequency RMSE and settling.

Dynamics
RoCoF tracking and sensitivity.

Deployability
CPU, latency, and jitter.

Study logic

Why problem	What literature gap	How method	Results measured	Insights deployment
field motivation	benchmark gap	experimental design	quantitative evidence	deployment guidance

> The argument moves from grid events to estimator qualification: define the stress, run matched estimators, evaluate multiple metrics, and map the result to deployment regimes.

1. Research Problem

Event-aware frequency metrics for low-inertia grids.

Research problem

- Low-inertia IBR grids require reliable local frequency estimates during ramps, phase jumps, harmonic distortion, ringdown, and mixed dynamic events.
- Classical estimators often trade off responsiveness, noise rejection, latency, and computational load.
- Modern model-based and data-driven estimators improve some regimes, but can be difficult to compare fairly.
- Existing comparisons often under-specify seeds, disturbance families, tuning policy, trip-risk, structural latency, or computational cost.

Research gap

The literature needs a reproducible benchmark that connects estimator error to protection-oriented risk, settling behavior, and deployability.

Research questions

1. Which estimator families lead each disturbance family: ramps, magnitude steps, modulation, phase jumps, OOB interference, harmonics, ringdown, and composite IBR events?
2. When do RMSE, peak error, trip-risk duration, and settling time select different estimators?
3. How does CPU cost change a ranking that appears favorable under accuracy alone?
4. Which scenarios are hardest, and what do they reveal about IBR-oriented protection measurement?
5. How do severity sweeps change the interpretation of the five-scenario benchmark?

> The central result is conditional selection: there is no universal estimator; the defensible choice depends on disturbance family, objective metric, and compute budget.

2. Benchmark Gap

Standards, datasets, estimator evidence.

Benchmark literature gap

IEC/IEEE PMU standards → NIST metrology → Open PMU platforms → Real PMU datasets → Algorithm studies

- **IEC/IEEE PMU standards:** Compliance for synchrophasor, frequency, and ROCOF measurements
- **NIST metrology:** Traceable calibration and PMU performance assessment
- **Open PMU platforms:** Device and streaming infrastructure for PMU research
- **Real PMU datasets:** Operational events from real grids
- **Algorithm studies:** Deep studies of selected estimator families

- The field is mature in standards, metrology, data collection, and individual estimator studies.
- The missing layer is a reproducible, estimator-level benchmark that compares many algorithms under identical synthetic truth and deployment metrics.

Source: IEC/IEEE 60255-118-1:2018; NISTIR 8106; OpenPMU; openPDC; FNET/GridEye; LBNL Open micro-PMU; PNNL open PMU event library

Benchmark gaps in prior work

Prior benchmark type	Strength	Gap for estimator-family comparison
IEC/IEEE PMU compliance	Defines synchrophasor, frequency, ROCOF, static/dynamic compliance tests	Leaves algorithm choice, family comparison, and CPU/latency outside the estimator report.
NIST PMU assessment/calibration	Provides metrological evaluation of PMU devices against standard requirements	Focuses on instruments and compliance behavior rather than a large matrix of open estimator implementations under controlled tuning policy.
OpenPMU / openPDC ecosystem	Enables open PMU construction and synchrophasor stream processing	Infrastructure-oriented, with no controlled estimator matrix using matched seeds and stress sweeps.
Open real PMU datasets	Supply realistic events for detection and application development	Truth is partially latent; hard to isolate causal stress factors or compute estimator error against known $f_{\mathrm{true}}[k]$.
Single-study algorithm comparisons	Give detailed evidence for selected methods or scenarios	Usually limited estimator sets, limited artifact contract, and incomplete deployability metrics.

Compliance tests define requirements, operational data define realism, and controlled synthetic truth isolates estimator behavior under reproducible stress.

Benchmark contribution

Known-truth scenario factory: steps, ramps, modulation, harmonics, interharmonics, noise

↓

Estimator execution contract: common API, matched seeds, warm-up, fixed/oracle policy

↓

Metric vector: RMSE, RFE, trip risk, settling, CPU, latency, jitter, invalid outputs

↓

Traceability: CSV/JSON artifacts, manifests, readiness gates, regenerated figures

↓

defensible estimator selection

The experimental system explains why estimator rankings change across objectives and regimes.

Benchmark comparison axes

Evidence source	Known truth	IBR composite stress	Trip-risk	CPU/latency	Reproducible artifacts
IEC/IEEE-style compliance	Yes	Limited	No	No	Device-dependent
Field PMU/DFR data	No	Yes	Possible	No	Dataset-dependent
Single-method studies	Often	Partial	Partial	Partial	Often limited
Estimator benchmark	Yes	Yes	Yes	Yes	Yes

Measured outputs

The same voltage trace is scored by RMSE, peak error, RoCoF error, trip exposure, settling, CPU, and structural latency.

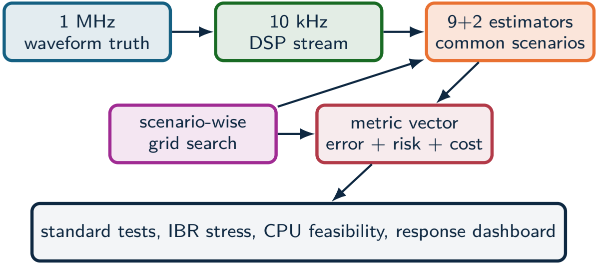
Interpretive effect

An estimator can pass standard cases and still fail a composite IBR event; it can also be accurate but too delayed or costly for protection-adjacent use.

3. Experimental Method

Signals, scenarios, I/O, metrics.

Experimental method



Source: Methodology and test framework

Measurement setting

Local, non-time-synchronized relay-class frequency tracking from a decimated 10 kHz stream; PMU timestamp certification is outside scope.

Experimental principle

Each estimator–scenario pair is tuned within a declared grid, then compared using RMSE, peak error, trip exposure, settling, and CPU.

RA-EKF mechanism

State augmentation

$$x_k = [\theta_k, \omega_k, A_k, \dot{\omega}_k]^\top, \qquad \omega = 2\pi f$$

$$\theta_k = \theta_{k-1} + \omega_{k-1}\Delta t + \frac{1}{2}\dot{\omega}_{k-1}\Delta t^2$$

$$\omega_k = \omega_{k-1} + \dot{\omega}_{k-1}\Delta t$$

IBR-specific stress response

$$Q_k = \mathrm{clip}(r_k, 0.25, 4)\, Q_{\mathrm{base}}, \qquad r_k = \frac{|\nu_k|}{\hat{\sigma}_\nu}$$

- **Covariance scaling**: inflate process noise under transients, deflate near steady state.
- **Event gating**: when $|\nu_k| > \gamma\hat{\sigma}_\nu$, softly reset θ_k while preserving ω_k and $\dot{\omega}_k$.

RA-EKF extends the EKF baseline with an explicit RoCoF state and event-aware covariance behavior for phase-jump recovery and steady-state tracking.

Metrics and decision variables

Error and recovery

$$e[k] = \widehat{f}[k] - f_{\mathrm{true}}[k]$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_k e[k]^2}$$

$$E_{\max} = \max_k |e[k]|$$

Protection-oriented risk

$$T_{\mathrm{trip}} = \Delta t \sum_k \mathbf{1}\{|e[k]| > 0.5\ \mathrm{Hz}\}$$

$$t_s = \min\{t : \ |e[\tau]| \le \epsilon,\ \forall \tau \ge t\}$$

$$C_{\mathrm{CPU}} = \mathrm{mean\ time/sample}$$

The 0.5 Hz threshold is a protection-oriented risk proxy. The metric set separates average error, transient severity, trip exposure, settling, and compute cost.

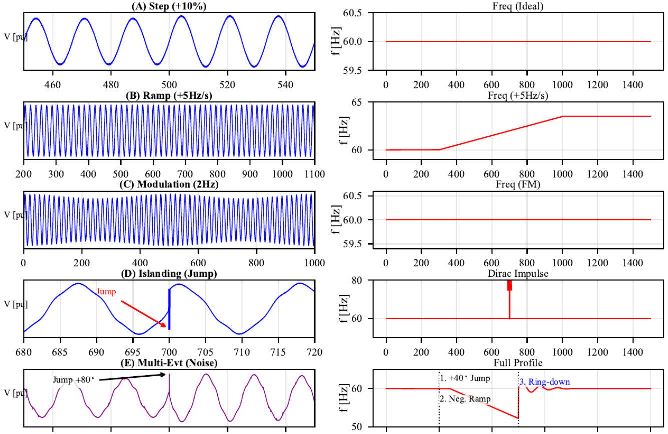
Scenario design: five stress cases

Case	Disturbance	Purpose
A	$+10\%$ voltage magnitude step.	Standard-style amplitude decoupling test.
B	$+5$ Hz/s frequency ramp.	RoCoF tracking and lag under sustained acceleration.
C	10% AM modulation at 2 Hz.	Steady-state modulation and noise sensitivity.
D	Composite islanding: $+60^\circ$ phase jump with harmonics/interharmonic and elevated noise.	Phase-jump recovery and false-trip exposure beyond standard tests.
E	IBR multi-event: harmonics, $+40^\circ$ jump, -6 Hz/s ramp, $+80^\circ$ jump, ringdown, impulsive noise.	Stacked stress resembling local IBR event records.

The design deliberately separates standard-style cases (A–C) from IBR-specific composite stress (D–E), which is where compliance pass/fail alone becomes inadequate.

Source: Scenario definitions and Fig. 1

Scenario suite



Scenario structure

- A–C are standard-style references.
- D introduces phase discontinuity and harmonic stress.
- E stacks the event mechanisms that standards isolate.
- The figure explains why RMSE, T_{trip}, and CPU must be read together.

Estimator families

Loop-based

SRF-PLL, SOGI-FLL.

Window-based

IpDFT, TFT.

Model-based / stochastic

EKF, UKF, RA-EKF.

Adaptive and data-driven

VFF-RLS, Koopman (RK-DPMU), PI-GRU.

Legacy baselines

RLS and TKEO are retained as comparison references outside the recommended deployment set.

Interpretation rule

These families occupy different latency and stress-sensitivity positions. The comparison tracks where each family wins, degrades, and fails.

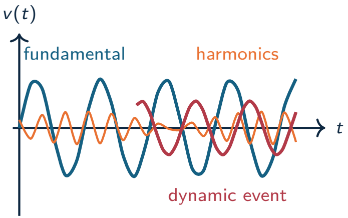
Signal and disturbance model

Observed single-phase voltage

$$v[k] = A[k]\sin(\theta[k] + \phi_0) + \sum_{h\in\mathcal{H}} a_h[k]\sin(h\theta[k] + \phi_h) + \eta[k]$$

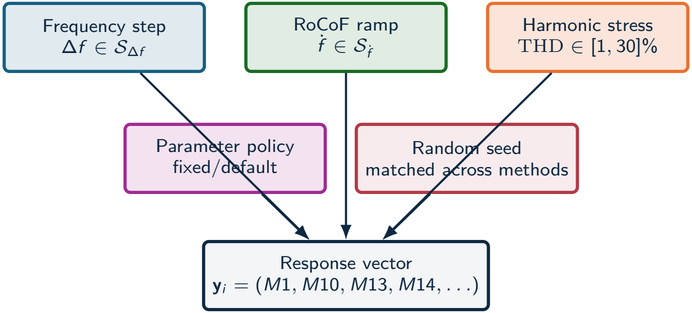
$$f_{\mathrm{true}}[k] = \frac{1}{2\pi}\frac{d\theta(t)}{dt}\bigg|_{t=k/f_s}$$

$$\mathrm{RoCoF}[k] = \frac{df_{\mathrm{true}}(t)}{dt}\bigg|_{t=k/f_s}$$



Known $f_{\mathrm{true}}[k]$ lets each estimator be tested under explicit violations of ideal sinusoidal assumptions.

Experimental design factors



- The design is factorial in stress type, stress level, estimator family, policy, and Monte Carlo seed.
- The response vector keeps accuracy, recovery, latency, and cost as separate outputs.

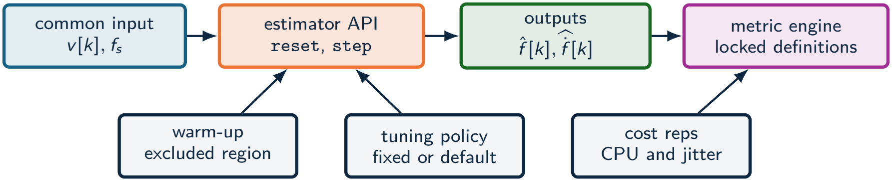
Benchmark methodology

Scenario $v[k]$, $f_{\text{true}}[k]$ → Monte Carlo matched seeds → Estimator common API → Locked metrics M1, M10, M13 → Artifacts CSV/JSON/PDF

- Common sampling profile: $f_s = 10$ kHz under `canonical-single-phase-v1`.
- Metrics are defined in code and documentation; scenario configuration is separate from metric definitions.
- Every result traces back to reports, manifests, CSV tables, and benchmark metadata.

Source: `docs/METHODS.md`

Estimator execution contract



- Fairness is enforced through identical input traces, matched random seeds, common warm-up handling, and a uniform output interface.
- Computational performance is measured as a response variable for deployability.

Metric equations

Frequency and derivative error

$$\mathrm{RMSE}_i = \sqrt{\frac{1}{|\mathcal{T}|}\sum_{k\in\mathcal{T}}\left(\hat{f}_i[k] - f_{\mathrm{true}}[k]\right)^2}$$

$$\mathrm{RFE}_i = \sqrt{\frac{1}{|\mathcal{T}|}\sum_{k\in\mathcal{T}}\left(\widehat{\dot{f}}_i[k] - \dot{f}_{\mathrm{true}}[k]\right)^2}$$

Deployability vector

$$\mathbf{J}_i = [\mathrm{RMSE}_i,\ \mathrm{RFE}_i,\ \mathrm{CPU}_i,\ \tau_i,\ \mathrm{TripRisk}_i]$$

$$i \prec j \iff \mathbf{J}_i \le \mathbf{J}_j \quad \text{and} \quad \exists m : J_{i,m} < J_{j,m}$$

The Pareto relation is the correct formal object when no single estimator dominates all scientific and deployment dimensions.

Experimental evidence matrix

Core benchmark evidence

- 9 principal estimators plus 2 legacy baselines.
- 5 scenarios: A–C standard-style, D–E IBR-specific.
- Standard, islanding, multi-event, and CPU results use the same scenario definitions and metric engine.
- Main results: RA-EKF transient performance, latency/stress trade-off, and compliance-test limits.

Representative runs with 18 estimators

- `frequency_step_mvp2 _all18_representative`
- `freq_ramp_rocof_mvp2 _all18_representative`
- Policy: `fixed_policy`; $n = 1$ Monte Carlo.

ATLAS P0 harmonic pilot

- `atlas-harmonics-fast14-preview-v2`
- 14 estimators, THD range 1–30%.
- Policy: `default`; $n = 1$ Monte Carlo.

ATLAS role

- Show mechanisms behind the aggregate rankings.
- Explain sensitivity curves and readiness gates.
- Identify which sweeps need deeper replication.

The core benchmark supports the main empirical conclusions; ATLAS pilot sweeps explain mechanisms and prioritize the next replication runs.

Estimator taxonomy

Loop-based

SRF-PLL, SOGI-FLL.

Window-based

IpDFT, TFT.

Model-based

EKF, UKF, and **RA-EKF** with explicit $\dot{\omega}$ state and event gating.

Adaptive

VFF-RLS plus RLS/TKEO legacy baselines.

Data-driven

Koopman (RK-DPMU), PI-GRU.

Rationale

Family-level analysis separates algorithmic principles from implementation-specific parameter choices.

Metric contract

- **M1 RMSE**: frequency tracking error.
- **M5 trip risk**: time outside protection deadband.
- **M8 settling**: post-event recovery.
- **M9/M10 RFE**: RoCoF tracking error.
- **M13 CPU**: per-sample compute cost.
- **M14 latency**: declared structural delay.
- **M20 jitter**: execution-time variability.
- **M31–M33**: estimates pinned to bounds.

Frequency accuracy, derivative accuracy, latency, and computational cost are reported as separate evidence dimensions.

Source: `docs/METHODS.md`

Evidence-level gate

Observed status: diagnostic

↓

$n = 1$ MC, 2/18 estimators

↓

policy: default

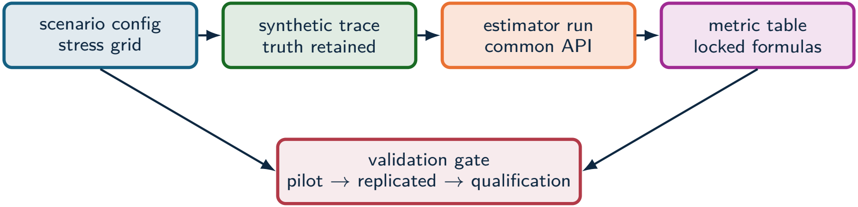
↓

target: fixed policy $n \geq 100$, full ATLAS

- ATLAS smoke status: **diagnostic**.
- Reported readiness issues: **10**.
- Full benchmark evidence requires complete sweeps, the canonical estimator set, fixed policy, and at least $n \geq 30$.
- High-confidence qualification should repeat the same protocol with $n \geq 100$.

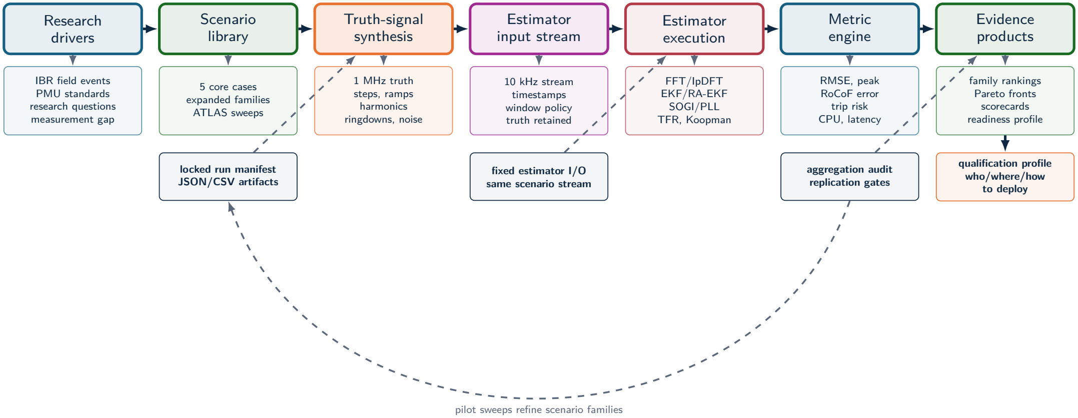
Source: `artifacts/atlas-readiness-smoke/atlas_readiness_report.json`

Experimental process and traceability



- Each figure is derived from machine-readable CSV/JSON outputs generated by the same experimental pipeline.
- Traceability is part of the method: scenario definition, estimator outputs, metric tables, and readiness status are archived together.

Full benchmark workflow



4. Benchmark Evidence

Standard cases, IBR stress, CPU cost, and deployment limits.

Standard scenario results

Method	Step +10% RMSE	Step +10% Peak	Ramp +5 Hz/s RMSE	Ramp +5 Hz/s Peak	Modulation 2 Hz RMSE	Modulation 2 Hz Peak
IpDFT	0.00157	0.00741	0.0729	0.157	0.000281	0.000747
TFT	0.00210	0.0190	0.0726	0.127	0.000260	0.000781
Koopman	**0.000207**	0.00523	0.0549	0.0860	**0.000212**	0.00178
SRF-PLL	0.000261	0.00132	0.142	0.354	0.000238	**0.000441**
SOGI-FLL	0.00900	0.0100	2.12	3.05	0.00900	0.0100
EKF	0.000733	0.00857	0.0180	0.0631	0.00119	0.00265
UKF	0.000774	0.00851	0.0207	0.0657	0.00116	0.00271
RA-EKF	0.00231	0.0299	**0.0113**	**0.0441**	0.00476	0.0125

Key result: RA-EKF is the strongest ramp tracker in this table, with 0.0113 Hz RMSE, 12.6× below SRF-PLL and 1.59× below EKF.

Source: `Table III`

Composite islanding: 275× trip-exposure reduction

Scenario D evidence

- Instantaneous $+60^\circ$ phase jump with harmonic and interharmonic content.
- Windowed leakage: IpDFT reaches 0.481 Hz RMSE; TFT reaches 2.74 Hz.
- EKF/UKF incur $T_{\mathrm{trip}} = 0.165$ s without phase-reset mechanisms.
- SRF-PLL is competitive on RMSE, 0.0645 Hz, with $T_{\mathrm{trip}} = 0$.

RA-EKF result

$$\mathrm{RMSE} = 0.0695\ \mathrm{Hz}$$

$$T_{\mathrm{trip}} = 0.6\ \mathrm{ms}$$

$$0.165\ \mathrm{s}/0.0006\ \mathrm{s} \approx 275\times$$

The gain comes from explicit $\dot{\omega}$ modeling plus event gating rather than post-hoc filtering.

Protection reading: RA-EKF converts long EKF false-trip exposure into a sub-millisecond transient under the islanding phase jump.

Multi-event trip-risk

Method	RMSE [Hz]	T_{trip} [s]	CPU [μs]
RA-EKF	0.511	0.141	54
EKF	0.789	**0.118**	18
Koopman	**0.476**	0.137	85
UKF	0.792	0.119	139
SRF-PLL	0.400	0.471	11
IpDFT	1.36	0.335	48
SOGI-FLL	0.568	0.422	25

Main reading

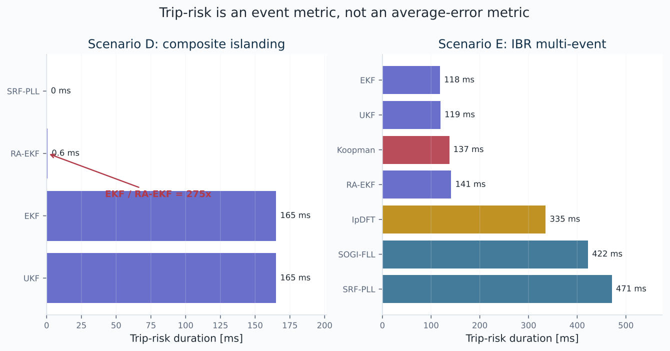
RA-EKF sits in the practical protection region: low trip exposure, moderate RMSE, and embedded-feasible CPU.

Scalar ranking failure

SRF-PLL has low CPU and good RMSE but much higher T_{trip}; EKF has lower T_{trip} but worse RMSE; Koopman is accurate but costlier.

Source: `Table IV`

Trip-risk evidence: core stress cases



Result reading

- Scenario D exposes phase-reset behavior: RA-EKF reduces EKF exposure from 165 ms to 0.6 ms.
- Scenario E is multi-objective: RMSE and trip-risk select different methods.
- SRF-PLL is strong in Scenario D but reaches 471 ms in Scenario E.

Result: T_{trip} separates protection exposure from average error.

Result comparison: winners depend on the question

Question	Best-supported answer	Evidence	Boundary
Clean step/modulation accuracy	Koopman and SRF-PLL are strongest point cases	Step RMSE 0.000207 Hz; modulation peak 0.000441 Hz	Not enough for islanding or multi-event survival
Ramp and RoCoF tracking	RA-EKF is strongest in the standard ramp	Ramp RMSE 0.0113 Hz	Not a universal RMSE winner
Composite islanding exposure	RA-EKF removes long EKF exposure	0.6 ms vs 165 ms for EKF	SRF-PLL is also strong in this specific phase-jump case
Multi-event RMSE	Koopman and SRF-PLL are competitive	0.476 Hz and 0.400 Hz	RMSE hides trip-risk and compute trade-offs
Embedded protection feasibility	SRF-PLL, EKF, SOGI-FLL, RA-EKF remain plausible	11–54 μs/sample range	Python timing is a software-cost proxy

The result is a conditional selection rule: choose by event family, objective metric, and compute envelope. A scalar leaderboard would misrepresent the benchmark.

Computational feasibility

Method	Time/sample	Class	Target
SRF-PLL	11 μs	P2	MCU
EKF	18 μs	P1	DSP
TFT	25 μs	P1	DSP
SOGI-FLL	25 μs	P2	MCU/DSP
IpDFT	48 μs	M1	DSP/FPGA
RA-EKF	**54 μs**	**P1**	**DSP/FPGA**
Koopman	85 μs	M1	CPU
UKF	139 μs	P1	high-perf CPU
PI-GRU	$\approx$136,000 μs	–	GPU only

Source: `Table V`

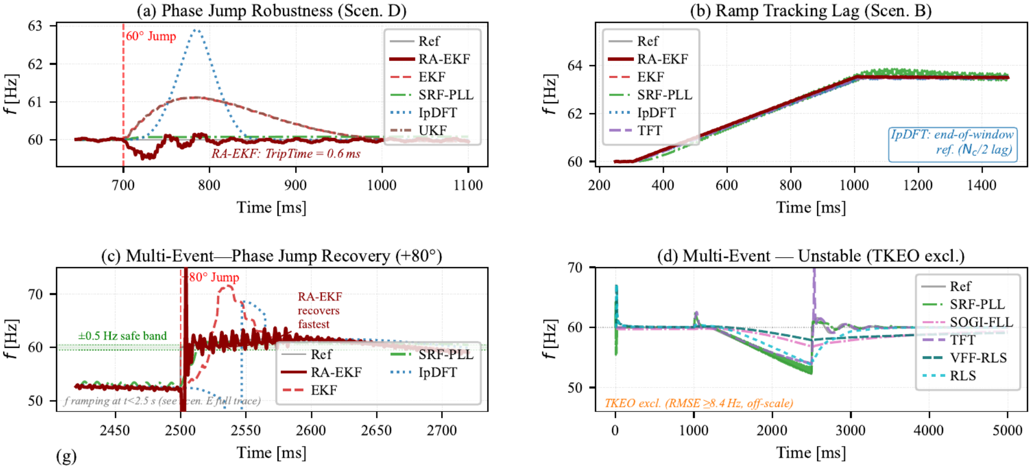
Deployment interpretation

The Python baseline separates feasible estimator families from clearly prohibitive ones; hardware latency requires platform timing.

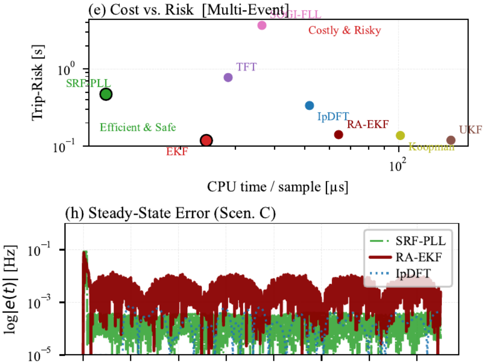
Key comparison

RA-EKF is roughly three times EKF cost in Python but remains below the millisecond scale by a wide margin; PI-GRU is retained as a computational reference outside embedded protection targets.

Event responses



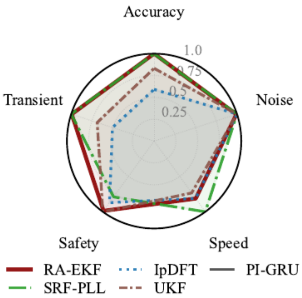
Deployment limits



Dashboard reading

- Deployment risk requires RMSE, trip exposure, settling, and delay.
- Trip-risk exposes the cost of estimator transients in composite events.
- Windowed methods can be accurate but delayed.
- RA-EKF occupies the main protection-oriented trade-off point.

Balance profile



Radar profile

- Accuracy, speed, noise immunity, safety, and transient behavior are conflicting axes.
- RA-EKF is selected for event-rich protection use cases.
- SRF-PLL remains valuable when cost and simplicity dominate.
- IpDFT is attractive where observation delay is acceptable.

Compliance heatmap: why standard tests are insufficient

Method	Step	Ramp	Mod	Isl	Multi
PI-GRU	F	F	F	F	F
Koopman	P	F	P	F	F
TFT	P	F	P	F	F
IpDFT	P	F	P	F	F
SOGI-FLL	P	F	P	P	F
SRF-PLL	P	F	P	F	F
EKF	P	P	P	F	F
UKF	P	P	P	F	F
RA-EKF	P	P	P	F*	F

Several methods pass the standard-style Step/Ramp/Mod columns and still fail under composite islanding or multi-event IBR stress. Standard tests leave IBR event readiness partially untested.

High-value empirical claims

Protection behavior

RA-EKF turns EKF exposure from long to sub-cycle.
Scenario D gives $T_{\mathrm{trip}} = 0.6$ ms for RA-EKF versus 0.165 s for EKF, while preserving an explicit dynamic state model.

Metric disagreement

RMSE, trip-risk, and CPU select different winners.
Table IV separates Koopman/SRF-PLL accuracy, EKF/UKF trip exposure, and low-cost loop/filter deployment.

Standard-test limitation

Simple-case pass/fail leaves event readiness unresolved.
The heatmap shows methods that pass Step/Ramp/Mod but fail islanding or multi-event IBR stress.

Deployability

Feasibility must be scored with accuracy.
RA-EKF remains in a 54 μs/sample software-cost regime; PI-GRU is retained as a computational reference outside relay-class timing.

Guardrail: report scenario, metric, and compute envelope before naming a preferred estimator.

Composite IBR sequence: why it is hardest

Scenario E stress components

- Harmonic-rich steady state: 5th at 5%, 7th at 3%, low interharmonics.
- $+40^\circ$ phase jump.
- -6 Hz/s RoCoF segment.
- $+80^\circ$ phase jump with ringdown.
- Sparse impulsive noise.

Standard-test gap

- The estimator sees a nonstationary waveform with stacked disturbances.
- Spectral leakage, loop re-lock, state-model mismatch, and recovery delay occur together.
- A method can be good on RMSE and still undesirable on trip exposure or CPU.

Once composite stress is shown to matter, ATLAS sweeps each stress component systematically instead of treating the multi-event case as a single aggregate disturbance.

Expanded stress analysis

EXPANDED BENCHMARK: 16 estimators, 34 scenarios, 60 Monte Carlo runs per scenario

Core experiment

- 9 principal estimators + 2 legacy baselines.
- 5 scenarios: A–E.
- Standard, islanding, multi-event, and CPU results.
- RA-EKF supplies the tested estimator mechanism.

Expanded benchmark

- 16-estimator, 34-scenario, 10-family benchmark.
- 60 Monte Carlo runs per scenario in the expanded dataset.
- ATLAS severity curves and harmonic sweeps.
- Used to test sensitivity, scaling, and estimator-family stability.

The expanded benchmark scales the same measurement question across more event families, estimator types, and Monte Carlo replications.

Expanded benchmark: family winners

EXPANDED BENCHMARK: 16 estimators, 34 scenarios, 60 Monte Carlo runs per scenario

Family	Count	RMSE leader	RMSE	Trip-risk leader
IEEE ramps	9	SOGI-PLL	19.4 mHz	0 s, multiple estimators
Magnitude steps	7	EKF	0.388 mHz	0 s, multiple estimators
Modulation	3	RA-EKF	17.8 mHz	0 s, multiple estimators
Phase jumps	3	SOGI-FLL	174 mHz	ZCD, 0.0194 s
OOB interference	1	EKF	13.6 mHz	EKF / RA-EKF, 0 s
IBR harmonics	3	EKF	70.5 mHz	TFT / Prony, 0.00947 s
IBR ringdown	5	UKF	313 mHz	ESPRIT, 0.0989 s
IBR multi-event	1	PLL	173.5 mHz	IpDFT, 0.0439 s

The expanded sweep confirms the central pattern: estimator rankings depend on disturbance family and metric target.

ATLAS severity analysis

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy

Core cases → ATLAS severity curves → sign asymmetry → fixed vs oracle policy → readiness gate

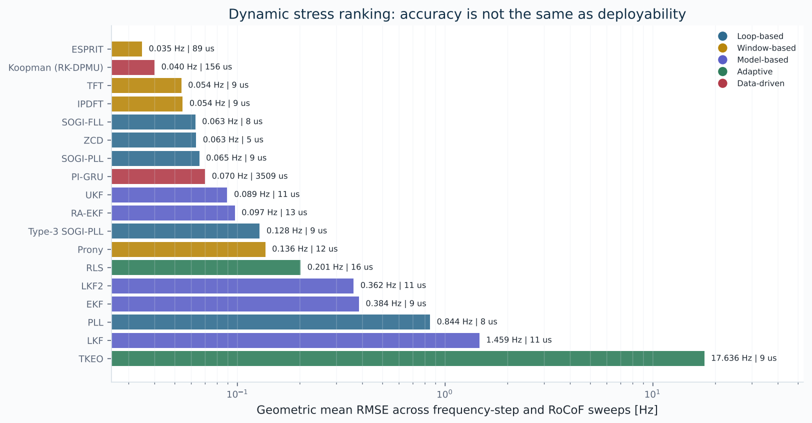
Core cases	ATLAS severity curves	sign asymmetry	fixed vs oracle policy	readiness gate
A–E failure mode	When does error start to grow?	Are positive and negative events symmetric?	Does tuning policy change the rank?	Which results replicate?

ATLAS maps where estimator behavior changes with severity, sign, tuning policy, and replication depth.

Source: `docs/ATLAS_PIPELINE.md`; `docs/ATLAS_METHOD_AUDIT.md`

Dynamic accuracy ranking

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Frequency RMSE: leading estimators

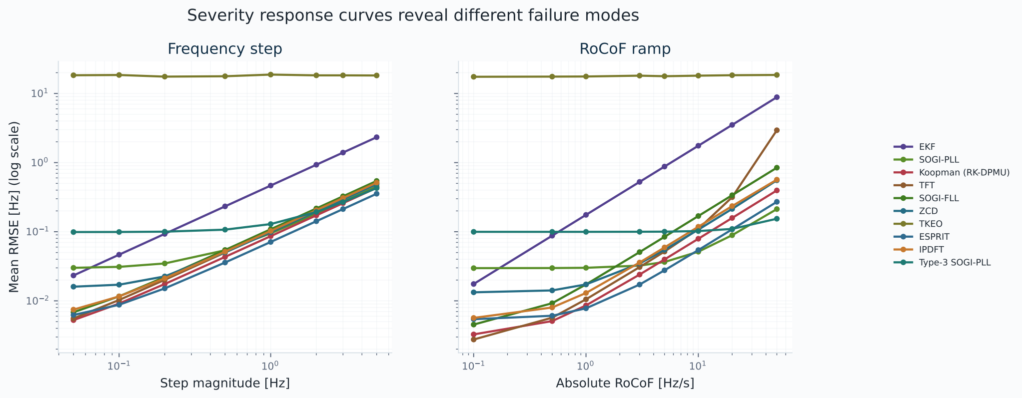
ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy

Estimator	Family	Geo. RMSE	Worst	CPU us	Latency ms
ESPRIT	Window-based	0.035	0.355	89	8.3
Koopman (RK-DPMU)	Data-driven	0.04	0.432	156	12.5
TFT	Window-based	0.054	2.952	9	16.6
IpDFT	Window-based	0.054	0.59	9	33.3
SOGI-FLL	Loop-based	0.063	0.844	8	0
ZCD	Loop-based	0.063	0.65	5	16.7
SOGI-PLL	Loop-based	0.065	0.457	9	0

- ESPRIT and Koopman provide the lowest geometric dynamic RMSE in the analyzed runs.
- TFT, IpDFT, SOGI-FLL, ZCD, and SOGI-PLL form a competitive low-cost tier.
- PI-GRU is accurate but its CPU cost changes its deployment interpretation.

Sensitivity to disturbance severity

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Interpreting sensitivity curves

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy

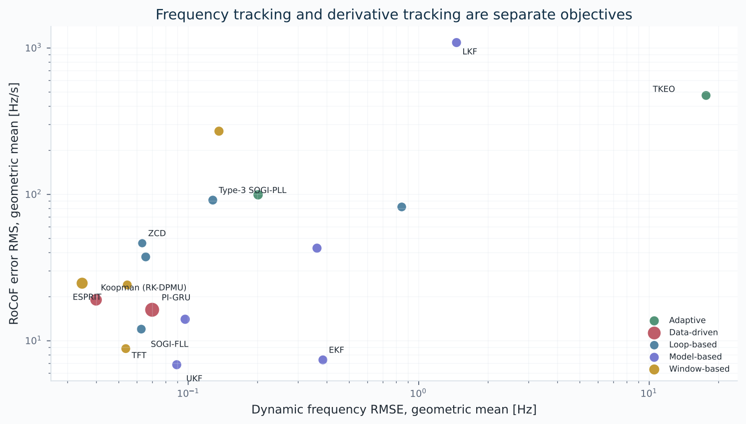
- Power-law-like growth indicates a measurable stress response.
- A nearly flat curve can still mean high baseline error.
- Type-3 SOGI-PLL has weak average frequency RMSE but favorable worst-case behavior under selected RoCoF conditions.

Stress	Estimator	Slope	Ratio	Sign asym.
RoCoF	LKF	0.086	2.1	5.52
RoCoF	TKEO	0.01	1.1	4.83
RoCoF	Type-3 SOGI-PLL	0.049	1.5	1.18
Step	LKF	0.017	1.1	2.29
Step	TKEO	1.33e-03	1	1.54
Step	EKF	1	100.1	1.01

Classify curves as stable-flat, failed-flat, monotone-sensitive, or sign-asymmetric before interpreting aggregate error.

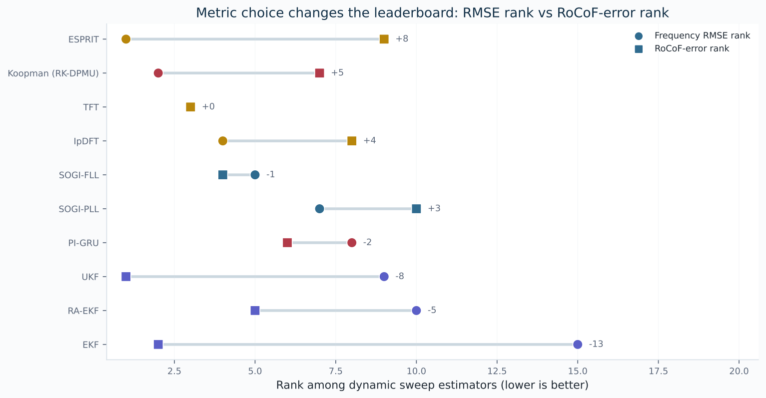
Frequency and RoCoF disagree

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Rank shift when the metric changes

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Derivative tracking: top RoCoF results

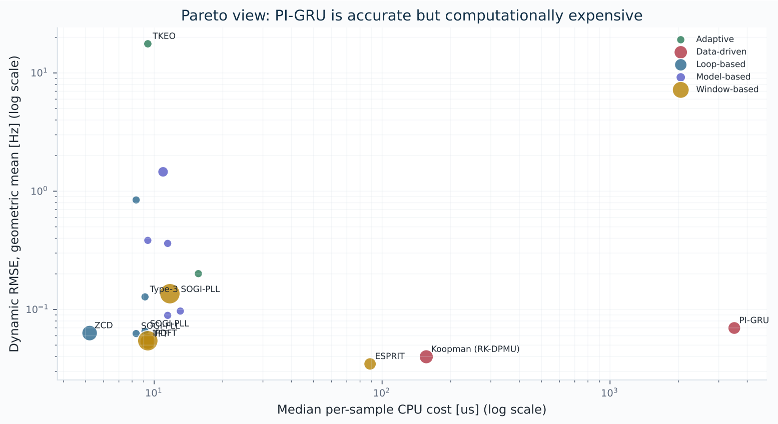
ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy

Estimator	Family	Geo. RFE	Geo. RMSE	CPU us	Latency ms
UKF	Model-based	6.864	0.089	11	0
EKF	Model-based	7.406	0.384	9	0
TFT	Window-based	8.842	0.054	9	16.6
SOGI-FLL	Loop-based	12.017	0.063	8	0
RA-EKF	Model-based	14.033	0.097	13	0
PI-GRU	Data-driven	16.293	0.07	3509	9.1
Koopman (RK-DPMU)	Data-driven	19.046	0.04	156	12.5

- UKF, EKF, TFT, SOGI-FLL, and RA-EKF move upward when the metric is RoCoF error.
- ESPRIT and Koopman remain competitive, but they no longer dominate the ranking.
- RoCoF error needs its own ranking because derivative performance has a separate order from frequency RMSE.

CPU–accuracy Pareto analysis

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Interpreting the Pareto front

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy

Low-cost online monitoring

ZCD, SOGI-FLL, SOGI-PLL, TFT, and IpDFT sit close to the low-cost frontier, with window latency as the main caveat for TFT/IpDFT.

Data-driven estimation

PI-GRU is empirically competitive in RMSE but requires a hardware and batching argument because of CPU cost.

High dynamic accuracy

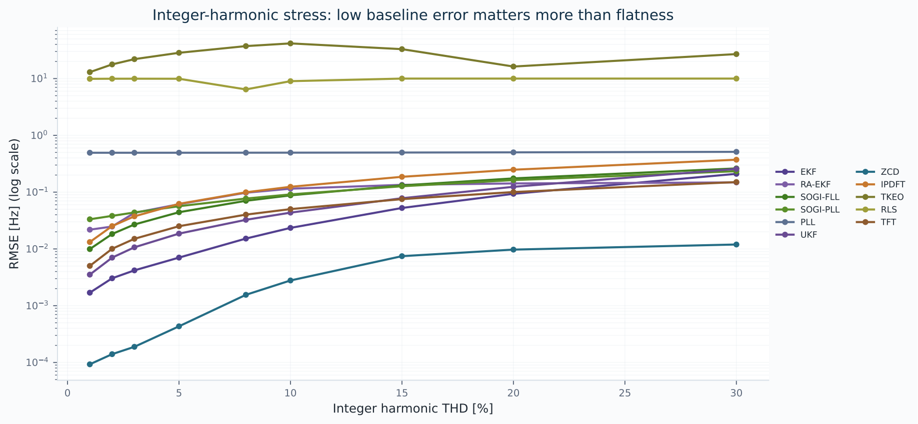
ESPRIT and Koopman are strong candidates when additional compute and latency are acceptable.

Methodological consequence

A single leaderboard would hide deployment constraints that matter in grid instrumentation.

Harmonic-stress response

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy



Harmonic sweep interpretation

ATLAS SEVERITY SWEEP: pilot run, $n = 1$, default controller policy

Estimator	Family	1% THD	30% THD	Ratio 30/1
ZCD	Loop-based	9.29e-05	0.012	128.787
RA-EKF	Model-based	0.022	0.148	6.797
TFT	Window-based	5.02e-03	0.15	29.95
EKF	Model-based	1.69e-03	0.21	123.981
SOGI-PLL	Loop-based	0.033	0.234	7.029
UKF	Model-based	3.53e-03	0.256	72.649
SOGI-FLL	Loop-based	9.93e-03	0.262	26.344

- ZCD is unexpectedly strong under isolated integer-harmonic stress in the pilot harmonic run.
- Generalization to interharmonics, noise, IBR-rich events, or mixed disturbances requires additional runs.
- RA-EKF, TFT, EKF, SOGI-PLL, and UKF remain secondary candidates for harmonic stress.
- Low slope must be read with baseline RMSE; both values are needed.

5. Insights and Validity

Rankings, recommendations, validity.

Three findings

1. **Latency and stress trade-off**: IpDFT/TFT provide strong steady-state accuracy but carry 3–5 cycle structural delay; loop methods are fast but can suffer elevated trip exposure.
2. **State-space estimation matters**: RA-EKF's explicit $\dot{\omega}$ state plus event gating reduces Scenario-D trip exposure from 0.165 s for EKF to about 0.6 ms.
3. **Compliance tests are insufficient**: estimators that pass standard-style Step/Ramp/Mod cases can fail under composite islanding and multi-event IBR stress.

> Estimator qualification includes transient recovery, trip exposure, compute feasibility, and standard-case RMSE.

Validity limits and next steps

Validation limits

- Limited to the reported scenario set.
- CPU is a software-level cost indicator; hardware latency requires platform timing.
- The study is simulation-only.
- Scenario-wise tuning can affect rank stability.

Validation program

- Hardware-in-the-loop validation.
- Wider converter-control interaction coverage.
- Sensitivity analysis of the tuning protocol.
- ATLAS severity and sign sweeps with readiness gates.

These limits define the validation path: broader event coverage, hardware timing, fixed-policy replication, and field replay.

Methodological findings

1. Disturbance family changes the estimator ranking and must be the first axis of interpretation.
2. RMSE, peak error, trip-risk duration, and settling time should be evaluated as separate objectives.
3. Latency and CPU cost are scientific response variables for deployability.
4. Composite IBR stress is necessary because isolated compliance-style cases hide mixed-event failure modes.
5. ATLAS readiness gates prevent pilot sweeps from being interpreted as final qualification evidence.

> Controlled traces, locked metrics, and artifact manifests make estimator comparisons reproducible.

Evidence-based recommendation map

FFR / RoCoF tracking	Anti-islanding phase jumps	Low-cost relay core	Metering-like steady state
↓	↓	↓	↓
RA-EKF explicit $\dot{\omega}$	RA-EKF event gating	SRF-PLL / EKF CPU efficient	IpDFT / TFT if delay acceptable

Deployability depends on event class, trip-risk tolerance, and compute budget.

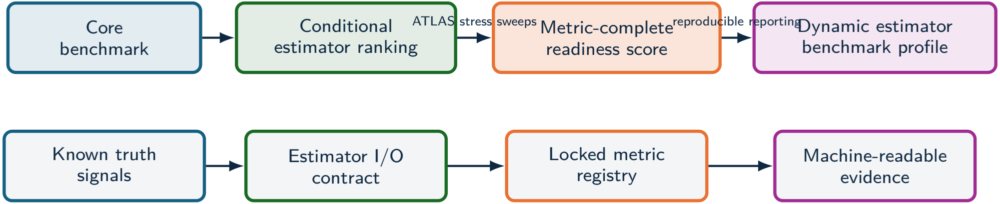
Four results

- **RA-EKF targets phase-jump recovery**: explicit $\dot{\omega}$, covariance scaling, and event gating reduce false-trip exposure.
- **Trip-risk separates from RMSE**: Scenario D and E show why cumulative time outside a protection band must be reported separately.
- **CPU changes the answer**: SRF-PLL and EKF remain strong low-cost references; PI-GRU falls outside protection-real-time in the reported baseline.
- **ATLAS expands the stress space**: severity, sign, policy, latency, and reproducibility analyses explain ranking stability.

6. Evidence Profile

Qualification logic and next validation.

Benchmark evidence profile



Estimator selection moves toward **qualification** when each method is tested against a declared event profile, metric profile, and latency-cost envelope.

Standardization profile

Current standardization layer

- PMU standards define phasor, frequency, and RoCoF measurement requirements.
- Compliance tests are necessary, with estimator-family ranking by IBR event survival left outside scope.
- Device reports often hide algorithmic failure modes behind aggregate accuracy.

Standardization boundary

A dynamic-estimator profile needs declared signals, estimator I/O, metrics, evidence artifacts, and latency-cost reporting.

Benchmark profile

- Canonical events: steps, ramps, phase jumps, ringdown, harmonics, interharmonics, noise, and composite IBR sequences.
- Locked metrics: RMSE, peak error, RoCoF error, trip-risk, settling, invalid samples, CPU, and latency.
- Task profiles: forensic monitoring, PMU augmentation, DFR support, and relay-adjacent screening.

Standardization value

A reference test profile needs the same signals, estimator I/O, metrics, and machine-readable evidence.

IBR Event Readiness Score

EVENT READINESS METRIC: weighted event-class qualification score

Score for estimator e

$$\text{IBR-ERS}(e) = 1 - \sum_{s\in\mathcal{S}} w_s\, \Phi_s(e),$$

$$\Phi_s(e) = \alpha\tilde{E}_s + \beta\tilde{P}_s + \gamma\tilde{T}_s + \delta\tilde{S}_s + \eta\tilde{C}_s + \lambda\tilde{L}_s.$$

- $\tilde{E}_s$: normalized dynamic RMSE.
- $\tilde{P}_s$: normalized peak deviation.
- $\tilde{T}_s$: trip-risk duration.
- $\tilde{S}_s$: settling and recovery penalty.
- $\tilde{C}_s, \tilde{L}_s$: compute and latency penalties.

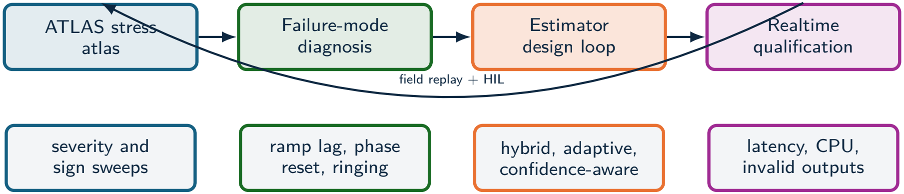
Composite metric rationale

The results show that RMSE, trip-risk, transient recovery, and low-cost candidates lead to different decisions. A readiness score makes the deployment objective explicit instead of hiding it inside a single leaderboard.

Profile-specific weights

Protection-adjacent use can raise γ, δ, λ; forensic reconstruction can raise α, β; embedded monitoring can raise η.

Toward realtime IBR estimators



- The stress analysis identifies which physical stressor breaks each estimator family.
- That diagnosis supports hybrid estimators: event classifiers, harmonic-aware preprocessing, adaptive bandwidth, and self-reported confidence.
- Validation should combine synthetic truth, EMT replay, hardware-in-the-loop, and field oscillography.

Validation agenda

Near-term extensions

- Full ATLAS sweeps with fixed policy and 100-run Monte Carlo replication.
- Statistical rank stability under tuning, noise, and disturbance severity.
- Composite readiness profiles for forensic, PMU-augmentation, and relay-adjacent tasks.

Longer-term research program

- Three-phase, unbalance, and converter-control interaction profiles.
- Real-event replay from DFR/PMU/oscillography with synchronized ground-truth proxies.
- Estimators that optimize an objective vector: frequency error, RoCoF, trip exposure, settling, CPU, and latency.

The broader goal is a reproducible science of **dynamic frequency observability**, where estimators are evaluated by the event regimes they can survive in real time.

Full-scale experimental protocol

Target command

```
python -m pipelines.atlas_sweep -sweeps all
-policy fixed_policy -n-runs 100 -n-cost-reps
3 -tune-trials 80 -output-subdir
atlas-full-v1
```

Source: `docs/ATLAS_PIPELINE.md`

- Complete canonical estimator set.
- Full ATLAS sweep suite.
- Archived seeds, manifests, and hashes.
- Predefined hypotheses and figure scripts.
- Regenerate all tables from CSV/JSON artifacts.

Conclusion

- Field disturbances define the measurement problem; controlled synthetic truth makes estimator behavior explainable.
- The key contribution is controlled comparison across objectives, regimes, and deployment constraints, including IBR-relevant stress modes.
- Empirical conclusions remain tied to evidence level, stress regime, and replication depth.

Result

The framework selects frequency estimators before they enter forensic, PMU, or protection-adjacent workflows.

Selected benchmark references

- IEEE/IEC 60255-118-1:2018: synchrophasor, frequency, and ROCOF measurement definitions and compliance requirements.
- NISTIR 8106: NIST assessment of PMU performance against synchrophasor requirements.
- NIST PMU calibration service: traceable calibration of PMUs against IEEE synchrophasor standard requirements.
- OpenPMU: open-source PMU device/platform for phasor measurement research.
- openPDC: open-source phasor data concentrator for streaming synchrophasor infrastructure.
- FNET/GridEye: wide-area frequency disturbance recorder network for grid monitoring.
- LBNL Open micro-PMU datasets and PNNL open-source PMU event library: real event data for algorithm and application development.
- NERC recommended IBR disturbance monitoring: high-resolution DFR data, continuous PMU/DDR data, and inverter fault-code retention for post-mortem analysis.

> These references motivate the gap: standards and datasets leave room for controlled, estimator-level tests with known truth and deployability metrics.

Selected field-event references

- NERC/WECC Blue Cut Fire report: 1,200 MW fault-induced solar PV resource interruption, Southern California, August 16, 2016.
- NERC/WECC Canyon 2 Fire report: 900 MW-class fault-induced solar PV resource interruption, Southern California, October 9, 2017.
- NERC Odessa disturbance reports: widespread solar PV reductions in Texas and associated model-fidelity and inverter-control findings.
- NERC 2022 California BESS disturbance report: abnormal BESS power reductions following normally cleared Western Interconnection faults.
- NERC Synchronized Measurements Subcommittee white paper on recommended disturbance monitoring for IBRs.
- Dong et al., Kaua'i 2021 18–20 Hz oscillation analysis: IBR-driven subsynchronous oscillation diagnosis and EMT replay.
- Ofgem/National Grid ESO 9 August 2019 GB outage reports and ENTSO-E 28 April 2025 Iberian incident report: grid-wide cascading events where fast dynamic visibility is central to diagnosis.

> The field cases define stress modes. No claim is made that a single estimator would have prevented a disturbance.

Technical questions

Question	Short answer
Why scenario-wise tuning?	It estimates best-in-range behavior under the tested search space. Fixed-policy replication is the next qualification layer.
How is this different from IEC/IEEE compliance?	Compliance tests define device-level requirements. This study compares estimator algorithms under known truth, IBR-specific stress, trip-risk, and CPU cost.
Why Python CPU times?	CPU reports relative software cost. Embedded or FPGA timing requires implementation-specific replication.
What should be deployed?	For FFR and phase-jump anti-islanding, RA-EKF is the strongest benchmark-backed recommendation; SRF-PLL/EKF remain low-cost baselines; IpDFT/TFT are useful where observation delay is acceptable.
How reliable are ATLAS pilots?	They are pilot sweeps: $n = 1$, default policy. Final qualification requires fixed policy, full estimator coverage, and replication.
Does this explain any blackout?	No. Field events define the stress modes; no prevention claim is made.

Artifact sources

- artifacts/frequency_step_mvp2_all18_representative/global_metrics_report.csv
- artifacts/frequency_step_mvp2_all18_representative/frequency_step_hypothesis_tests.csv
- artifacts/freq_ramp_rocof_mvp2_all18_representative/global_metrics_report.csv
- artifacts/freq_ramp_rocof_mvp2_all18_representative/rocof_hypothesis_tests.csv
- artifacts/atlas-harmonics-fast14-preview-v2/global_metrics_report.csv
- artifacts/atlas-readiness-smoke/atlas_readiness_report.json
- slides/generate_slide_assets.py

Regeneration

Figures and tables

```
python slides\generate_slide_assets.py
```

PDF

```
cd slides
pdflatex slide.tex
pdflatex slide.tex
```